S10 CONTEXT BUDGET

Each base LLM runs inside the designated agent's CLI (either Codex or Claude Code), which manages the model's context. The CLI exposes file tools (listing, search, and reading) and runs them, appends the output to the conversation, and caps how much a single call returns. When the conversation approaches the model's context window, the CLI compacts it, and replaces the earlier turns with a summary.

The optimizer's history is stored as files in a read-only workspace. For each session, we take the workspace the optimizer was given on its last call and count its tokens with the `o200k_base` tokenizer. The saved workspaces range from 1.04M to 4.21M tokens, larger than any context window we use. The optimizer therefore never loads the history in full: it browses the workspace with these tools, and only the files it opens enter its context.

In Table S10.1, we show the detection call stats. Each detection call, unlike what the optimizer has, sends one harness and one function. For each code sample, the evaluation pipeline calls the harness's build function, which inserts the code into the harness's template. The detector then sends the resulting filled-in template to the LLM. The largest detection call was about 77K tokens, from a harness learned in the DeepSeek run. It fit easily within the windows of both models that ran it: 1.31M tokens for DeepSeek V4 Flash and 200K for GLM-4.7 Flash. Nemotron had the smallest window, which we set to 65.5K, and its largest call was only 22.5K. No detection call exceeded its model's window, so no harness or function was ever cut off.

Table S10.1: Context window and the size of one detection call (the harness plus one PrimeVul function), per model.

Model	Role	Agent	Context window	Per detection call (median / max)
Claude Opus 4.6	optimizer, detector	Claude Code	200K	3.4K / 23.7K
Claude Opus 5	detector (transfer)	Claude Code	200K	3.5K / 23.7K
GLM-5.3	optimizer, detector	Claude Code	200K[a]	5.8K / 27.8K
gpt-oss-120b	detector (transfer)	Claude Code	131K	5.8K / 27.8K
Inkling-Small	optimizer, detector	Codex CLI	256K	2.4K / 44.4K
Nemotron 3.5 Lightning	detector (transfer)	Codex CLI	65.5K[b]	2.5K / 22.5K
DeepSeek V4 Flash	optimizer, detector	Codex CLI	1.31M	5.7K / 76.9K
GLM-4.7 Flash	detector (transfer)	Codex CLI	200K	6.9K / 76.9K
GPT-4o	detector (transfer)	Codex CLI	128K	5.7K / 53.7K

[a] The model accepts 1.31M tokens; Claude Code runs it with the 200K window of the Claude model it is mapped to. [b] Set for Codex; the model accepts up to 1M. Token counts are estimated at 4 characters per token.